SELECTED INSTRUMENT / BENCH ACCEPTANCE PENDING

JMS TPM-02. Preserve the pressure trace.

Use the probe, connecting tube and manufacturer recording software as the proposed independent pressure reference. EMG is the model input; pressure is the label source.

Why this choice

The official software manual's CSV example contains measurement time and tongue pressure, in kPa, with time entries spaced 0.05 seconds apart. This provides a documented trace-export path for sustained holds and interruptions. The catalog lists USB data communication. These are manufacturer-document claims, not results from an Arc bench test.

> **Selected, not already validated.** No instrument has been purchased, connected or calibrated in this build. The example cadence does not guarantee every firmware/export configuration. No shared hardware synchronization interface was verified.

Import contract

PRESERVE	RULE
Original export	Retain immutable manufacturer CSV, instrument/firmware identity, calibration record and trial identity. Preserve original trial-relative timestamps.
Time mapping	Map each trial to the Arc session clock using independently measured offset/drift. Verify common synchronization events. Reject unmeasured alignment; never concatenate resetting trial clocks blindly.
Stored reference	reference_pressure, tongue_palate_pressure channel, kPa, explicit sample rate and per-sample validity. Preserve gaps; do not infer a hold through missing measurements.
Physical task	Record clinician-approved probe position, tongue-palate task, pressure criterion and meaningful interruption duration. Pressure verifies only this defined contact/force task, not every aspect of technique.

Implementation accepts aligned numeric arrays through the existing pack API. A direct device driver or manufacturer-specific CSV parser has not been fabricated from an example screenshot.

A label must reproduce from an independent trace.

PRESSURE RUN	STORED LABEL
Valid, at/above threshold, lasting at least hold_min_s	Positive maintained hold. Offline label spans the qualifying run from its original onset.
Valid, below threshold, lasting at least interruption_min_s	Negative interruption. Same onset convention.
Shorter run or invalid pressure	Exclude, never silently positive or negative. Invalid samples break runs.

The threshold and both durations are mandatory protocol inputs. One protocol decision rule must apply across the cohort. Per-session calibration and instrument identity remain individually recorded. Non-task periods require explicit exclusions.

What software now enforces

- Separate pressure stream, kPa units and declared alignment.
- Instrument, placement, calibration, source, protocol and approval fields.
- Protocol hash on generated labels; replayed pressure must reproduce scored labels.
- Train/validation pressure checked during development. Test pressure checked at held-out evaluation, never for threshold selection.
- Pressure channel excluded from runtime feature vectors.
- Incomplete and recovered sessions rejected from baseline training/evaluation.

What remains a real-world acceptance gate

The actual instrument, placement and calibration must be checked; export rate, omissions and clock alignment must be measured. A qualified clinician must approve task-specific numerical criteria before they are represented as approved. Software cannot authenticate those facts from a populated JSON field.

> **Bundled example is synthetic.** A generated pressure trace drives labels using 5 kPa and 0.1-second test values. Those constants exercise software; they are not suggested clinical thresholds. No human measurement or diagnostic performance is claimed.

Evidence bundle includes a loadable held-out session, channel/time/validity plot, exact CSV replay, baseline outputs and a SIGKILL recovery demonstration. Source scripts regenerate and verify those artifacts after extraction.